

CYPRESS HILLS PARK, SK, Canada

WMO: 711390

Lat: 49.6406N

Lon: 109.5144W

Elev: 4169

StdP: 12.61

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.6	-14.5	-26.0	1.5	-19.1	-20.4	2.1	-14.2	23.4	35.3	21.2	33.3	7.4	330	0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	83.2	58.9	79.9	58.3	76.3	57.6	62.8	75.3	60.9	74.0	59.2	72.4	8.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.6	85.7	67.5	56.1	78.3	65.6	54.0	72.5	63.7	30.5	75.0	29.0	74.3	27.7	72.3	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.0	17.7	15.8	-24.6	88.6	6.5	3.2	-29.3	90.9	-33.1	92.7	-36.8	94.5	-41.6	96.8		
			-24.8	66.5	6.4	2.7	-29.4	68.5	-33.1	70.0	-36.7	71.5	-41.3	73.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.9	17.5	18.1	26.3	37.6	47.7	55.2	62.8	62.0	52.7	39.7	27.4	18.7
	DBStd	19.93	16.00	14.51	13.61	10.13	8.69	6.88	6.43	7.38	9.71	10.32	12.65	14.37
	HDD50	5274	1009	893	734	385	148	26	2	5	80	342	679	971
	HDD65	9643	1474	1313	1199	822	538	300	121	146	383	784	1127	1436
	CDD50	1234	0	0	0	13	76	183	399	376	161	24	2	0
	CDD65	127	0	0	0	0	1	7	53	52	14	0	0	0
	CDH74	1234	0	0	0	1	20	77	492	508	135	1	0	0
	CDH80	268	0	0	0	0	1	10	109	127	21	0	0	0
Wind	WSAvg	7.8	8.4	7.9	8.4	8.4	8.2	7.3	6.9	6.8	7.3	8.2	8.1	8.2
Precipitation	PrecAvg	17.7	1.0	0.9	1.2	1.6	2.0	3.2	2.0	1.7	1.4	1.0	0.8	0.8
	PrecMax	28.2	2.9	1.7	2.3	3.5	5.1	10.3	3.8	5.2	2.8	5.0	1.9	1.8
	PrecMin	8.7	0.2	0.1	0.3	0.4	0.0	0.9	0.4	0.0	0.0	0.0	0.1	0.0
	PrecStd	4.2	0.7	0.5	0.6	0.8	1.5	2.3	1.2	1.3	0.9	1.1	0.5	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.7	49.7	56.6	68.6	76.9	81.4	87.0	87.6	83.1	70.0	57.7	49.6
		MCWB	38.7	38.3	43.3	49.0	54.1	59.9	60.9	59.3	56.6	51.8	44.1	37.3
	2%	DB	44.7	44.1	50.8	62.9	71.7	75.6	83.1	83.6	78.2	63.7	52.0	43.9
		MCWB	36.0	35.8	40.0	46.6	52.7	58.7	60.2	58.0	55.4	48.3	41.0	35.6
	5%	DB	41.0	39.9	46.8	57.7	67.0	71.5	79.7	80.0	73.2	58.4	47.6	40.4
		MCWB	33.9	33.1	37.7	43.7	50.5	56.7	59.7	57.7	53.9	45.8	38.6	33.7
	10%	DB	37.7	36.3	43.1	52.7	62.5	67.9	75.9	75.9	68.0	54.1	43.6	36.7
		MCWB	32.0	30.8	35.4	40.8	48.2	54.6	59.0	56.9	51.6	43.4	36.2	31.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.6	39.6	44.0	50.4	56.6	63.8	66.6	63.8	59.8	53.3	45.7	39.2
		MCDB	48.3	46.9	54.8	65.8	70.5	73.7	78.6	76.0	75.8	67.2	56.6	46.2
	2%	WB	36.7	36.3	41.0	47.2	53.8	60.4	63.8	61.6	57.1	49.4	41.8	36.2
		MCDB	42.9	43.2	49.1	60.7	68.5	71.6	76.5	75.1	73.4	61.0	50.4	42.9
	5%	WB	34.3	33.6	38.3	44.5	51.7	58.0	61.8	59.9	54.7	46.6	39.2	33.8
		MCDB	40.6	39.6	45.9	55.9	65.0	69.3	74.6	74.1	70.3	56.7	46.9	39.8
	10%	WB	32.0	30.8	35.7	41.8	49.3	55.7	59.9	58.3	52.5	44.0	36.4	31.2
		MCDB	37.5	36.0	42.8	51.7	60.3	66.1	72.7	72.5	66.5	53.1	42.9	36.6
Mean Daily Temperature Range	5% DB	MDBR	16.0	15.0	15.0	17.2	18.4	17.7	20.3	20.5	19.1	16.5	14.4	14.2
		MCDBR	16.6	17.6	19.4	24.3	24.5	22.4	24.9	25.6	25.0	22.2	18.3	16.4
		MCWBR	11.4	12.0	11.8	13.3	12.2	12.0	11.0	10.5	11.6	11.7	11.5	11.1
	5% WB	MCDBR	17.6	17.9	18.9	22.8	22.4	20.9	21.8	21.9	23.2	21.0	18.0	16.3
		MCWBR	13.0	13.0	12.1	13.1	11.9	12.1	11.4	10.7	11.6	11.6	12.0	11.8
Clear-Sky Solar Irradiance	taub	0.248	0.252	0.277	0.319	0.340	0.330	0.335	0.350	0.309	0.278	0.251	0.240	
	taud	2.410	2.454	2.478	2.381	2.362	2.448	2.442	2.374	2.465	2.538	2.508	2.411	
	Ebn at Noon	263	292	300	295	291	294	291	281	284	277	263	250	
	Edh at Noon	19	24	28	35	37	34	34	35	28	22	17	17	
All-Sky Solar Radiation	RadAvg	416	739	1159	1556	1850	1999	2176	1775	1281	800	457	322	
	RadStd	40	53	72	103	127	128	134	125	107	81	36	26	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-1.37	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page